R. Brown

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Department of Psychology

Dalhousie University

ABSTRACTS

1. G. V. Goddard, J. J. Miller and K. G. Baimbridge.

Biochemical Correlates of Kindling-Induced Epilepsy in the Hippocampus: The Role of Calcium Binding Protein (CaBP).

Rats were kindled to full motor convulsions by daily electrical stimulation of the hippocampal commissure. Radioimmunoassay revealed 25-30% reduction of CaBP in hippocampal tissue but not cerebellum or cerebral cortex and no change in calmodulin or total soluble protein. CaBP may regulate cellular discharge during strong synaptic activation.

2. R. S. Rodger.

A Classification and Comparison of Multiple Comparison Methods

Methods for evaluating multiple comparisons and multiple contrasts can be classified in a two-by-two table. One factor is whether the methods use decision-based error-rates or experiment-wise error-rates. The other factor is whether the comparisons or contrasts to be evaluated were planned (independently of the data used to evaluate them) or were selected post-hoc. The techniques t, Dunnett, Bonferroni, Duncan, Rodger, Scheffé, Tukey and Newman-Keuls fit neatly into this scheme. The factor identifications therefore provide a basis for choosing which method to use. It is relatively easy to depict graphically the increase in "harshness" of the methods as the number of groups (J) increases. For the popular value $\alpha = .05$, Scheffé's method becomes excessively harsh with increasing J while Tukey's half-versus-half is worse. Rodger's and Tukey's (comparisons only) increase their harshness moderately while Bonferroni's inequality (for J-1 contrasts), Dunnett's procedure and the Newman-Keuls method (on the average) are somewhat more generous. Duncan's procedure is the most generous of the post-hoc methods and Student's t is, of course, the most generous of all the methods. These facts are also relevant to a choice of methods. Finally, the various methods have limitations (to be illustrated) in what they can do and these also need to be taken into account when a method for analysis is selected.

3. P. Lynch and J. W. Clark.

Students' Opinions of Class Evaluations.

The present study was conducted to measure the relationship between students' attitudes of class evaluations and the overall rating of the quality of a class. An opinion questionnaire was given to students enrolled in three, second-level psychology classes. The findings were that: (1) the overall opinion of the three classes did not systematically vary with sex of student, sex of professor, majors or non-majors, or with year of study; (2) there is no evidence to suggest that the ratings of classes did not reflect in large measure the value of the classes.

4. D. B. Henken.

DNA Synthesis in the Goldfish Optic Tectal Explant.

An *in vitro* examination of the time course of cell proliferation in the periependymal strata of the optic tectum will be carried out. A pulse-chase labelling technique using $^3H$-thymidine will be employed to label newly synthesized DNA in dividing radial glial cells.

5. N. Bennett, Z. Havkin and J. Ryon.

Parental Behavior of Virgin Hand-Reared Timber Wolves (*Canis lupus*) on First Exposure to Young Pups.

Two female and one male four-year old virgin hand-reared timber wolves were individually introduced to a group of three pups between 48 and 52 days of age. The first regurgitation to the pups by each adult occurred within 3, 5, and 27 minutes, and all adults showed a wide range of parental behaviors within an hour. Although wolves are known to adopt alien young readily, the speed with which parental behavior was elicited from these adults, in addition to their complete lack of experience with pups or parents, makes these observations particularly striking. Indiscriminate attraction to pups is probably related to communal pup rearing in wolf packs.

6. M. Kaye, D. Mitchell and M. Cynader.

Loss of Binocular Depth Perception After Lesions of Cat Visual Cortex.

Three normal adult cats distinguished depth separations, equivalent to 4 minutes disparity, binocularly but not monocularly. Subsequent complete ablation of cortical areas 17 and 18 largely spared visual behaviour and acuity, but made even binocular depth discrimination substantially worse than pre-operative monocular discrimination. Visual cortex ablation selectively impairs depth perception.

7. M. Flynn, P. Dodd and W. K. Honig.

Differential Outcome Effects in a Delayed Conditional Discrimination.

Each trial began with red or green on the pigeon's key. This was followed by a retention interval (RI) when the key was dark. The test stimulus, a vertical or a horizontal line on the key, was presented next. On positive trials, the pigeon received food or water for pecking at the test stimulus. Negative trials ended with a blackout. Only sequences of red-vertical and green-horizontal were positive. For two birds the reinforced outcomes were differential, as all positive trials beginning with one color ended with grain, and all positive trials beginning with the other color ended with water. For two birds, these outcomes were non-differential, as food and water were randomly assigned to trials beginning with either color.

One short (1-sec) and one longer retention interval were presented in each session. The longer interval increased in the course of training. Performance at the 1-sec RI was similar for the two conditions; however, at the longer RI's, retention was better with the differential outcomes. This confirms recent research by Peterson and his associates who used a different form of delayed conditional matching. In their work, the proportion of non-reinforced trials depended on the subject's accuracy, while in the present study half the trials ended in blackout. Both studies indicate that different reward expectancies enhance the short-term retention of the initial stimulus.

8. I. A. Meinertzhagen and C. J. Arnett-Kibel.

Synaptic Connections of the Dragonfly Interneurons.

The synaptic connections of five classes of second-order interneuron of the visual system of the dragonfly *Sympetrum* have been analyzed from serial electronmicroscopy. Each is different; three are monochromatic while two have unique spectral and dichroic inputs.

9. E. Hansen and R. Klein.

Validity of Signal Locus Predictor and Relative Likelihood of Signal: When What You See Is Not What You Get.

Current work in our lab investigates effects of movement of attention in visual space (in the absence of eye movements) on subjects' detection of stimulus changes, using a reaction time cost-benefit paradigm. The experiments and preliminary data to be presented attempt to explain the absence of priming effects in low-probability stimulus-response sets. Two interpretations are possible: (1) attention is directed to stimulus attributes other than locus in a dual-stimulus situation; (2) attentional segregation occurs between response systems in a dual-response situation.

10. K. Grasse.

The Accessory Optic System in the Frog (*Rana Pipiens*)

The anatomy and physiology of the accessory optic system (AOS) has been investigated using single-unit extra-cellular recording and HRP-histochemistry. Units were evaluated electrophysiologically for their direction and speed selectivity.

Extremely small HRP injections (½ nanoliter or less) revealed a variety of novel anatomical features of the afferent and efferent organization of this system.

11. B. L. Bower and S. Nakajima.

Approach - Withdrawal Analysis of Median Raphe Stimulation in the Rat.

Rewarding stimulation of the median raphe nucleus reduced acceptability of sugar water, reduced aversiveness of a quinine solution, but did not influence the latency to escape from footshock. The results suggest that the nature of rewarding effect produced by raphe stimulation is different from that produced by hypothalamic stimulation.

12. R. Brown and D. Elrick.

Paternal Behaviour in Male Rats.

Adult rats which have never been exposed to infants will either kill or ignore them, but males who have cohabited with a female and her litter during lactation show low levels of pup-care. Paternal behaviour in male rats probably involves two processes: the inhibition of infanticide and the activation of pup-care behaviours. I will discuss data from three experiments which examine the effects of social experience on infanticide and pup-care in male rats. These experiments demonstrate that infanticide can be inhibited by social experiences with lactating females and pups and that male rats will show some paternal behaviour after exposure to rat pups.

13. L. Cauller.

Circadian Modulation of Hippocampal Excitability.

The magnitude of the granule cell potential evoked by a constant stimulus to the input fibres is greater during the night than the day. This circadian rhythm is evident when the data are grouped by behavioral/EEG state and contravaries with the responsiveness of the granule cells to the synaptic input. I am interested in the interaction of this rhythm with long-term potentiation of those synapses.

14. J. Barresi.

Perception and Knowledge.

A Kantian theory of the relationship between perceiving (e.g. seeing) and understanding (e.g. seeing as) is described and illustrated using examples from 'dot pattern' research.

15. T. Fauvel.

A Neuroethological Perspective on the Asymmetry of Gaze.

When a large part of the visual field moves around an animal, a sawtooth-like nystagmus of the eyes results. The strength of nystagmus appears to depend not only on the direction of motion in the field but also on where the field is in visual space. Ethological factors may have guided the evolution of the neural networks which underlie these asymmetries of gaze.

16. J. Radel.

Ultrastructure of the Goldfish Retino-tectal Synapse.

Electronmicrographs will be examined in an attempt to characterize (a) degenerating terminals of the optic fibers following enucleation, (b) the post synaptic sites of the primary visual neuron in the tectum, (c) normal presynaptic optic fiber terminals and (d) presynaptic terminals of regenerating optic fibers.

17. A. Fröhlich and I. A. Meinertzhagen.

Neurite Growth and Its Relationship to Synaptogenesis in the Fly's Visual System.

In the fly's visual system, the prospective postsynaptic neurites of the first interneurons grow to their presynaptic sites on receptor axons. We have examined this process in order to assess the influence of neurite growth on the formation of synapses.

18. M. Yoon and F. Baker.

Trophic Interaction Between Neurites of Retinal Tissue and Co-cultured Tectal Tissue of Adult Goldfish In Vitro.

The patterns of neuritic outgrowths from a retinal tissue into its topographically matching tectal tissue (explanted from adult goldfish) were studied in vitro under various experimental conditions. The tectal tissue tended to attract the retinal neurities to invade it. Within the tectal explant, the retinal neurites seemed to grow selectively towards the appropriate layers of their usual passage and termination in vivo.

19. D. Moore and I. A. Meinertzhagen.

<u>Synapses in the Fly's Visual System.</u>

The number of functionally appropriate synapses formed by a nerve cell is rarely defined. For identified photoreceptors of the fly, we provide evidence of the number of synapses, their constancy per cell, and their correlation with the area of presynaptic membrane (so providing constant membrane area per synapse).